Suggestions for Bio 3105/3107

Introduction

1. Why are you studying Biology?
2. How your knowledge will solve biological problem? Define a biological problem and solve using your knowledge (Any technologies)

Cell Structure

1. Cell Theory
2. Cell membrane (Fluid Mosaic Model)
3. Transport (Active and Passive); In what situations do active transporter and passive transporter transport solutes?
4. Organelles (Diagram- Mitochondria, Plastid, Golgi Body (Exocytosis))

Cell Division

1. Cell Cycle (Diagram)
2. Cell cycle regulators (indicate checkpoints in cell cycle)
3. Mitosis Vs Meiosis
4. (Prophase of Mitosis) Vs (Prophase I of Meiosis)
5. Genetic Variation (Diagram)

Genetics and Genomics

1. DNA and RNA (features)
2. Central Dogma / How DNA instructions can lead to form protein
3. DNA Replication (Diagram)
4. Codon/ Genetic Code (Definition and Characteristics)
5. Chromosomal abnormality (Flow chart, Aneuploidy, Down syndrome)
6. Karyotyping

Structural Biology

1. Definition
2. Protein structure

3. Cryo-EM technique